The Quantum Approximate Optimization Algorithm (QAOA)

Theory, Circuits, and Applications

Morten Hjorth-Jensen

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Outline

Motivation

Many problems in physics and computer science reduce to optimization:

- Ground-state energy minimization
- Spin models (Ising, Heisenberg)
- Combinatorial optimization (MaxCut, SAT)

General form:

$$\min_{z \in \{0,1\}^n} C(z)$$

QAOA is designed to solve such problems using quantum circuits.

Variational Quantum Algorithms

QAOA belongs to the class of hybrid quantum-classical algorithms.
Idea:

- Prepare a parametrized quantum state $|\psi(\boldsymbol{\theta})\rangle$
- Measure expectation value:

$$E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})| H_C |\psi(\boldsymbol{\theta})\rangle$$

- Optimize parameters classically

This is closely related to VQE.

Definition of QAOA

QAOA constructs the state

$$|\psi_p(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_C} |+\rangle^{\otimes n}$$

where:

- H_C: cost Hamiltonian
- H_M: mixing Hamiltonian
- p: circuit depth

Initial State

The algorithm starts from

$$|+\rangle^{\otimes n} = \frac{1}{2^{n/2}} \sum_{z} |z\rangle$$

This represents an equal superposition of all candidate solutions.

Cost Hamiltonian

The classical cost function is encoded as a diagonal Hamiltonian:

$$H_C = \sum_z C(z) \left|z\right\rangle \left\langle z\right|$$

Example (Ising form):

$$H_C = \sum_{i<j} J_{ij} Z_i Z_j + \sum_i h_i Z_i$$

Thus:

$$C(z) = \left\langle z\right| H_C \left|z\right\rangle$$

Mixing Hamiltonian

The standard choice is

$$H_M = \sum_{i=1}^{n} X_i$$

This generates transitions between computational basis states and ensures exploration of the search space.

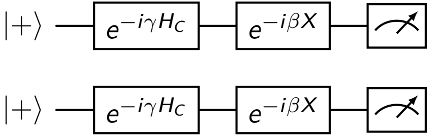
QAOA Circuit Structure

Each layer consists of:

1. Cost unitary:
$$U_C(\gamma) = e^{-i\gamma H_C}$$
2. Mixer:
$$U_M(\beta) = e^{-i\beta H_M}$$

Repeated p times.

QAOA Circuit (p=1)



Example: MaxCut Problem

Given a graph $G = (V, E)$, maximize:

$$C(z) = \sum_{(i,j) \in E} \frac{1 - z_i z_j}{2}$$

Mapping to Hamiltonian:

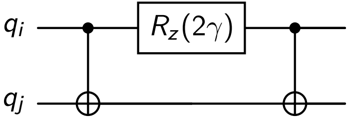
$$H_C = \sum_{(i,j) \in E} \frac{1 - Z_i Z_j}{2}$$

MaxCut Cost Unitary

For one edge:

$$e^{-i\gamma(1-Z_iZ_j)/2}$$

Circuit implementation:



Mixer Unitary

For each qubit:

$$e^{-i\beta X}$$

Circuit:

$$R_x(2\beta)$$

$$q_i \;-\!\!-\!\!\boxed{R_x(2\beta)}\!\!-\!\!-$$

Expectation Value

We evaluate:

$$\langle H_C \rangle = \langle \psi(\gamma, \beta)| H_C |\psi(\gamma, \beta)\rangle$$

This is estimated via measurements on the quantum device.

Classical Optimization Loop

1. Initialize parameters (γ, β)
2. Run quantum circuit
3. Measure expectation value
4. Update parameters using classical optimizer

Repeat until convergence.

Physical Interpretation

QAOA can be viewed as a discretized quantum annealing process:

$$H(t) = (1 - s(t))H_M + s(t)H_C$$

with alternating evolution under H_C and H_M.

Performance

- $p = 1$: shallow circuits, limited accuracy
- larger p: better approximation
- classical optimization becomes harder

Tradeoff:

$$\text{accuracy} \leftrightarrow \text{circuit depth}$$

Relation to VQE

QAOA is a special case of VQE:

- structured ansatz
- problem-inspired Hamiltonians

Difference:

- QAOA: alternating operators
- VQE: general parametrized circuits

Limitations

- Barren plateaus in optimization
- Hardware noise
- scaling with problem size

Summary

QAOA:

- encodes optimization problems into Hamiltonians
- uses alternating quantum evolution
- is hybrid quantum-classical
- connects quantum computing with statistical physics

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Outline

1. QAOA review
2. Adiabatic quantum computation
3. Trotterization $\Rightarrow$ QAOA
4. MaxCut as an Ising model
5. Analytical solution at $p = 1$
6. Statistical mechanics interpretation

QAOA State

$$|\psi_p(\gamma, \beta)\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_C} |+\rangle^{\otimes n}$$

- H_C: problem Hamiltonian
- $H_M = \sum_i X_i$: mixing Hamiltonian

Goal:

$$\max \langle\psi| H_C |\psi\rangle$$

Adiabatic Quantum Computation

Consider a time-dependent Hamiltonian:

$$H(s) = (1 - s)H_M + sH_C, \quad s \in [0, 1]$$

Adiabatic theorem:

- If evolution is slow, system stays in ground state
- Final state $\rightarrow$ ground state of H_C

Time evolution:

$$U = \mathcal{T} \exp\left(-i \int_0^T H(s)ds\right)$$

Discretization of Adiabatic Evolution

Divide time into p steps:

$$U \approx \prod_{k=1}^{p} e^{-i\Delta t((1-s_k)H_M+s_k H_C)}$$

Using Trotter decomposition:

$$e^{-i(A+B)\Delta t} \approx e^{-iA\Delta t}e^{-iB\Delta t}$$

From Adiabatic Evolution to QAOA

Applying Trotterization:

$$U \approx \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_C}$$

Thus:

$$\boxed{\text{QAOA is a discretized (Trotterized) adiabatic evolution}}$$

Key idea:

- Replace fixed schedule with variational parameters

MaxCut as an Ising Model

MaxCut cost:

$$C(z) = \sum_{(i,j)\in E} \frac{1 - z_i z_j}{2}$$

Hamiltonian form:

$$H_C = \sum_{(i,j)\in E} \frac{1 - Z_i Z_j}{2}$$

Equivalent to Ising model:

$$H = -\sum_{i<j} J_{ij} Z_i Z_j$$

Thus QAOA solves an Ising ground-state problem.

Statistical Mechanics Interpretation

Define partition function:

$$Z = \mathrm{Tr}(e^{-\beta H})$$

QAOA instead prepares:

$$|\psi(\gamma, \beta)\rangle \sim e^{-i\beta H_M} e^{-i\gamma H_C}$$

Interpretation:

- Imaginary time: classical Gibbs states
- Real time: quantum interference

QAOA interpolates between:

$$\text{thermal sampling} \leftrightarrow \text{coherent dynamics}$$

Analytical Example: Single Edge

Consider two qubits with one edge:

$$H_C = \frac{1 - Z_1 Z_2}{2}$$

Initial state:

$$|+\rangle^{\otimes 2}$$

Apply Cost Unitary

$$U_C(\gamma) = e^{-i\gamma H_C}$$

Since $Z_1 Z_2$ eigenvalues ± 1:

$$U_C = e^{-i\gamma/2} e^{i\gamma Z_1 Z_2/2}$$

Apply Mixer

$$U_M(\beta) = e^{-i\beta(X_1+X_2)}$$

Acts as rotations:

$$R_x(2\beta)$$

on each qubit.

Expectation Value

We compute:

$$\langle H_C \rangle = \frac{1}{2}(1 - \langle Z_1 Z_2 \rangle)$$

After algebra (standard result):

$$\langle Z_1 Z_2 \rangle = \sin(4\beta)\sin(2\gamma)$$

Optimization (p=1)

Thus:

$$\langle H_C \rangle = \frac{1}{2}(1 - \sin(4\beta)\sin(2\gamma))$$

Maximize by choosing:

$$\beta = \frac{\pi}{8}, \quad \gamma = \frac{\pi}{4}$$

This gives optimal cut value.

Interpretation of p=1 Result

Key insight:

- Even shallow circuits capture nontrivial correlations
- Performance improves with connectivity

For general graphs:

- expectation depends on local graph structure

Connection to Spin Systems

QAOA operates directly on spin Hamiltonians:

$$H = \sum J_{ij} Z_i Z_j$$

This is:

- classical Ising model
- quantum transverse-field Ising model

Thus QAOA $=$ controlled exploration of spin configurations.

QAOA vs Classical Methods

Classical:

- simulated annealing
- Monte Carlo

QAOA:

- uses interference instead of thermal sampling
- explores many configurations simultaneously

Summary

- QAOA arises from Trotterized adiabatic evolution
- Encodes optimization as Ising Hamiltonians
- p=1 case admits analytical solutions
- Connects quantum computing with statistical physics

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Overview

- Optimization problems and spin systems
- QAOA construction
- Adiabatic origin and Trotterization
- Analytical structure (p=1)
- General graphs and locality
- Landscape and optimization theory
- Statistical mechanics connections
- Control-theoretic interpretation

Optimization as Physics

Many optimization problems map to:

$$\min_z C(z)$$

Equivalent to finding ground state of:

$$H_C = \sum_z C(z) \left|z\right\rangle \left\langle z\right|$$

This is identical to:

- classical spin systems
- Ising Hamiltonians

Ising Representation

General Ising form:

$$H_C = \sum_{i<j} J_{ij} Z_i Z_j + \sum_i h_i Z_i$$

Binary variables:

$$z_i \in \{-1, 1\}$$

Thus QAOA operates directly on many-body Hamiltonians.

QAOA Ansatz

$$|\psi_p(\gamma, \beta)\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_C} |+\rangle^{\otimes n}$$

Mixer:

$$H_M = \sum_i X_i$$

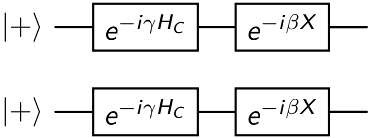
Physical Picture

- H_C: encodes energy landscape
- H_M: induces transitions

Thus:

$$\text{QAOA} = \text{controlled quantum exploration of configuration space}$$

Circuit Representation



Adiabatic Evolution

$$H(s) = (1 - s)H_M + sH_C$$

Evolution:

$$U = \mathcal{T} e^{-i \int_0^T H(s) ds}$$

Trotterization

Discretize:

$$U \approx \prod_{k=1}^{p} e^{-i\Delta t H(s_k)}$$

Apply Trotter:

$$e^{-i(A+B)} \approx e^{-iA} e^{-iB}$$

QAOA as Trotterized Adiabatic Evolution

$$\boxed{U \approx \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_C}}$$

Key difference:

- adiabatic: fixed schedule
- QAOA: variational parameters

MaxCut Problem

$$C(z) = \sum_{(i,j)} \frac{1 - z_i z_j}{2}$$

Hamiltonian:

$$H_C = \sum_{(i,j)} \frac{1 - Z_i Z_j}{2}$$

Local Structure

Important fact:

- At depth p, correlations extend only distance p

Thus expectation values depend only on local subgraphs.

Analytical p=1: Setup

Single edge:

$$H_C = \frac{1 - Z_1 Z_2}{2}$$

Initial state:

$$|+\rangle^{\otimes 2}$$

Evolution

State:

$$|\psi\rangle = e^{-i\beta H_M} e^{-i\gamma H_C} |{+}{+}\rangle$$

Key Computation

One finds:

$$\langle Z_1 Z_2 \rangle = \sin(4\beta)\sin(2\gamma)$$

Thus:

$$\langle H_C \rangle = \frac{1}{2}(1 - \sin(4\beta)\sin(2\gamma))$$

Optimal Parameters

Maximized at:

$$\beta = \frac{\pi}{8}, \quad \gamma = \frac{\pi}{4}$$

General Graphs

For arbitrary graph:

- expectation depends on local neighborhoods
- can be computed analytically for small p

Statistical Mechanics View

Define:

$$Z = \mathrm{Tr}(e^{-\beta H})$$

QAOA:

$$|\psi\rangle = e^{-i\beta H_M} e^{-i\gamma H_C} |+\rangle$$

Real vs Imaginary Time

- imaginary time: $e^{-\beta H} \to$ Gibbs states
- real time: $e^{-iHt} \to$ unitary evolution

QAOA operates in real time.

Relation to Annealing

- simulated annealing: thermal fluctuations
- quantum annealing: tunneling
- QAOA: controlled coherent evolution

Energy Landscape

Define:

$$E(\gamma, \beta) = \langle \psi | \, H_C \, | \psi \rangle$$

Properties:

- highly non-convex
- periodic

Parameter Landscape

- small p: smooth landscape
- large p: complex landscape

Connections to:

- spin-glass landscapes

Control-Theoretic View

QAOA:

$$U(\theta) = \mathcal{T} e^{-i \int dt [u_1(t) H_C + u_2(t) H_M]}$$

Piecewise constant controls:

$$u_1(t), u_2(t)$$

Continuous Limit

As $p \to \infty$:

$$\text{QAOA} \to \text{continuous Schrödinger evolution}$$

Connection to Optimal Control

Goal:

$$\max \langle\psi(T)| H_C |\psi(T)\rangle$$

This is a quantum control problem.

Relation to VQE

- QAOA: structured ansatz
- VQE: general ansatz

Scaling Challenges

- parameter optimization
- noise
- circuit depth

Barren Plateaus

Gradients may vanish exponentially:

$$\partial_\theta E \to 0$$

Why QAOA Still Works

Structure:

- locality
- problem-inspired ansatz

Summary

- QAOA $=$ Trotterized adiabatic evolution
- maps optimization $\rightarrow$ spin systems
- admits analytic results at small p
- connects quantum computing and statistical physics